



Pathlock Native Cyber Security and GRC Suite Dynamic Access Controls Installation Guide

Version: 2026 Q1



Pathlock

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1 Overview

SAP ECC/S4HANA establishes security through its **role-based access controls**, which assist organizations in preventing unauthorized access and mitigating risk violations.

Pathlock Dynamic Access Controls (DAC) functions as an additional security layer on the ECC/S4HANA application for SAP users. This add-on provides users with a granular approach to securing their ECC/S4HANA environments.

1.1 Purpose

This document provides high-level information regarding the installation of Pathlock DAC.

2 Pathlock DAC Installation

2.1 Transport Import Instructions

Pathlock DAC installation is based on deployment scenarios. Customers should select the appropriate option depending on their system landscape and requirements.

The following deployment options are supported:

- Option A: SAP GUI Only
- Option B: SAP GUI + UI5/Fiori
- Option C: SAP GUI + UI5/Fiori + BW

All required co-files and data files are available in the corresponding folders under the “Release Co-files” directory at the shared location.

Option A: SAP GUI Only

Folder Name: GUI_Only

Sequence	Folder Name	Transport Number
1	Workbench	YS9K900871
4	Roles	YS9K900885

Option B: SAP GUI + UI5/Fiori*Folder Name: GUI_UI5_Fiori*

Sequence	Folder Name	Transport Number
1	Workbench	D91K900455
2	Roles	YS9K900885
3	Workbench_Frontend	D91K900461
4	Roles_Frontend	D91K900435

Option C: SAP GUI + UI5/Fiori + BW*Folder Name: GUI_UI5_Fiori_BW*

Sequence	Folder Name	Transport Number
1	Workbench	D91K900457
2	Roles	YS9K900885
3	Workbench_Frontend	D91K900461
4	Roles_Frontend	D91K900435

Note:

- Ensure the RZ11 profile parameter "dynp/usrmasking" value is set to "OFF" before importing co-files into the system, including target systems such as quality or production environments.
- Import only the transport relevant to your selected scenario.
- Always import Workbench transport before Roles (Customizing).
- Workbench_Frontend transports must be imported into the Fiori frontend system (SAP_UI). In embedded deployments, import them into the backend system.
- The system must be on SAP ECC 6.0 EHP7 or higher to import the Workbench transports.
- To import BW Workbench transports, the system must be on SAP S/4HANA 1709 or higher with SAP_BW 7.52 or above installed.

2.2 Pathlock DAC Un-Installation

The following table outlines the steps, in the given sequence, to uninstall Pathlock DAC.

Step	Instruction
1	Deactivate RZ11 profile parameter "dynp/usrmasking" by setting its value to OFF.
2	Remove SPIs implemented as part of section 3 & 4 of this document.
3	Remove DAC API calls from the system if in case API scenarios are implemented.
4	Import cofiles available in the "Uninstallation" folder.

Sequence	Folder Name	Transport Number
1	Uninstallation	SFRK900046

2.3 Transport Notes

Use the transport order mentioned in section 2.1 to import the software into the SAP system. Save the transport files in the package to the following SAP directories:

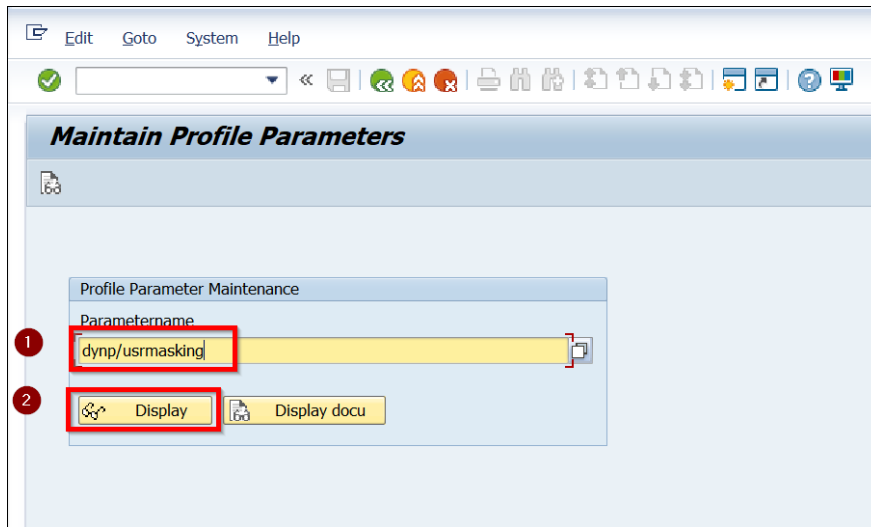
- "D" & "R" files: Save to `/usr/sap/trans/data`
- "K" files: Save to `/usr/sap/trans/cofiles`

Note: If any import errors with return code 8 occur, download the transport error log and email it to support@pathlock.com. Warnings with return code 4 can be ignored.

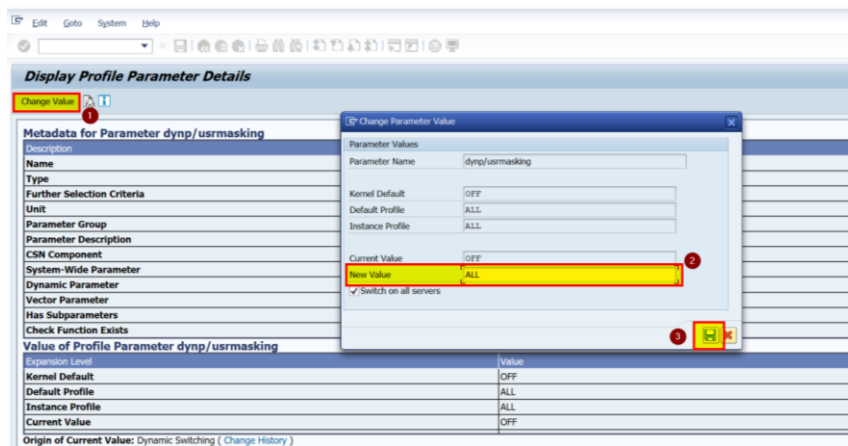
2.4 Profile Parameter Activation/Deactivation

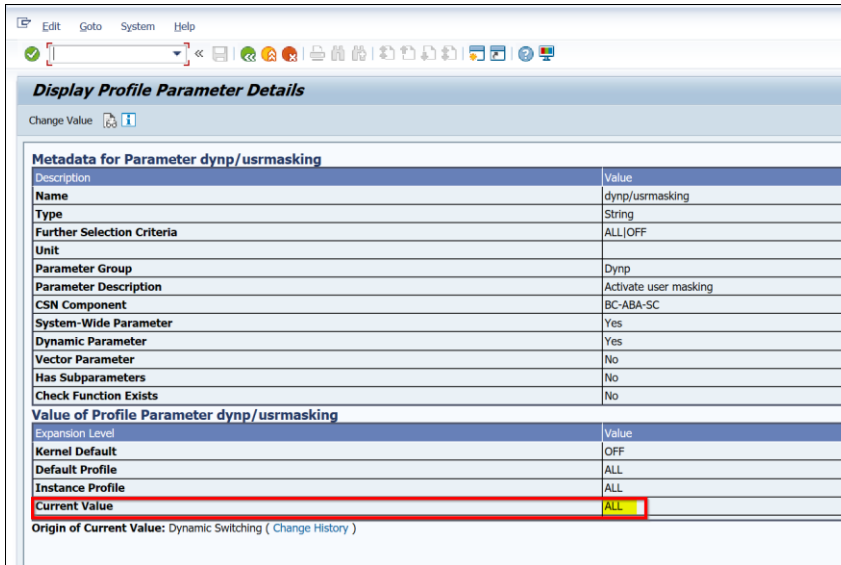
To activate the profile parameter "**dynp/usrmasking**," follow these steps:

- Go to transaction code **RZ11**.
- Provide the parameter "**dynp/usrmasking**" and click "Display."



If the current value is 'OFF', you can change the current value by selecting the 'Change Value' option.





Display Profile Parameter Details

Change Value

Metadata for Parameter dynp/usrmasking	
Description	Value
Name	dyp/usrmasking
Type	String
Further Selection Criteria	ALL OFF
Unit	
Parameter Group	Dyp
Parameter Description	Activate user masking
CSN Component	BC-ABA-SC
System-Wide Parameter	Yes
Dynamic Parameter	Yes
Vector Parameter	No
Has Subparameters	No
Check Function Exists	No

Value of Profile Parameter dynp/usrmasking	
Expansion Level	Value
Kernel Default	OFF
Default Profile	ALL
Instance Profile	ALL
Current Value	ALL

Origin of Current Value: Dynamic Switching ([Change History](#))

Ensure the profile parameter's current value remains 'ALL' to activate the Pathlock DAC solution system wide. To deactivate the parameter, set its value to 'OFF' to deactivate Pathlock DAC system wide.

Note: Changes in the parameter value are valid only as long as the application server is running. When the application server restarts, the values from the profiles (kernel, default, or instance profile) are reread and reset. To ensure the Pathlock DAC solution remains active upon system restart, set the parameter's current value to 'ALL' in the default profile.

3 SAP GUI (Enhancement / Modification)

Note: Ensure the profile parameter's current value remains 'OFF' until the SPI implementation is complete.

Create the single point of interception (SPI) via enhancement/modification as described in the following steps:

3.1 SAP GUI: Module Pool

Transaction code	SE38
Program	SAPFS_SECLOG
Subroutine	CALL_SECLOG_HOOK

Create the SPI in the aforementioned program at the start of the subroutine by copying the code snippet below.

Snippet:

```

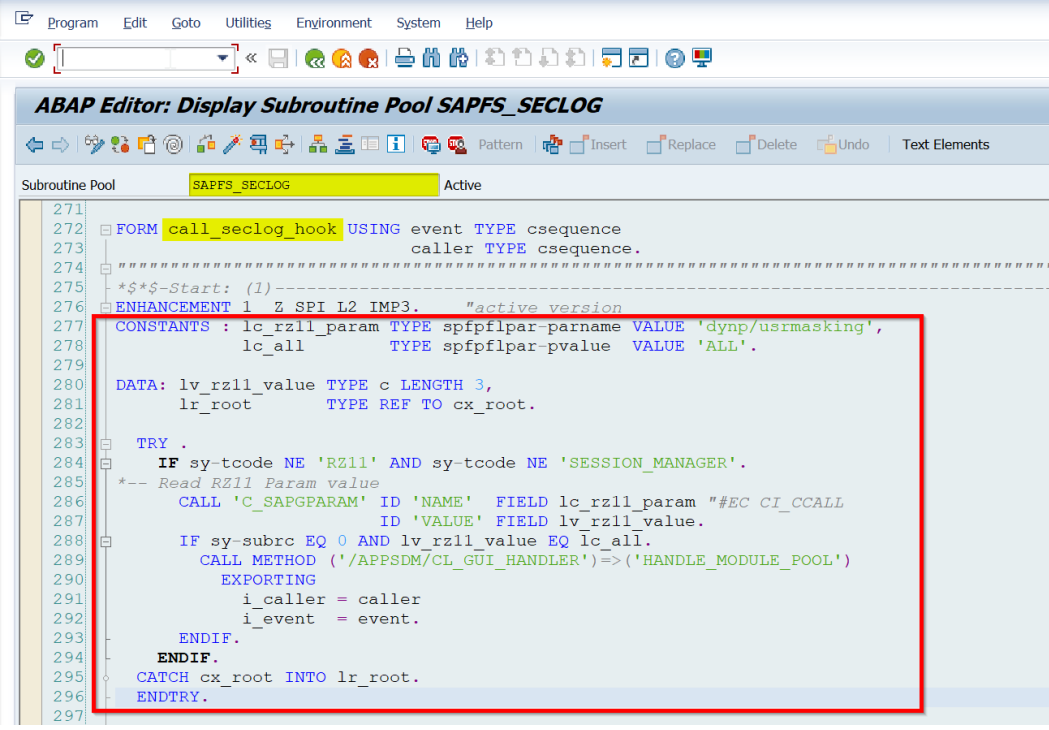
CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all        TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_rz11_value TYPE c LENGTH 3,
      lr_root       TYPE REF TO cx_root.

TRY .
  IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
*-- Read RZ11 Param value
    CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
                                ID 'VALUE' FIELD lv_rz11_value.
    IF sy-subrc EQ 0 AND lv_rz11_value EQ lc_all.
      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_MODULE_POOL')
        EXPORTING
          i_caller = caller
          i_event  = event.
    ENDIF.
  ENDIF.
CATCH cx_root INTO lr_root.
ENDTRY.

```

Screenshot:



```

271
272 FORM call_seclog_hook USING event TYPE csequence
273      caller TYPE csequence.
274
275 *****Start: (1)-----
276 ENHANCEMENT 1 Z SPI L2 IMP3. "active version
277 CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usimasking',
278      lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
279
280 DATA: lv_rz11_value TYPE c LENGTH 3,
281      lr_root TYPE REF TO cx_root.
282
283 TRY .
284     IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
285     *-- Read RZ11 Param value
286     CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
287     ID 'VALUE' FIELD lv_rz11_value.
288     IF sy-subrc EQ 0 AND lv_rz11_value EQ lc_all.
289     CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_MODULE_POOL')
290     EXPORTING
291         i_caller = caller
292         i_event = event.
293     ENDIF.
294 ENDIF.
295 CATCH cx_root INTO lr_root.
296 ENDTRY.
297

```

Note: Perform a syntax check before activating the code. Activating code with syntax errors can lead to system-wide runtime errors.

3.2 SAP GUI: ALV Grid

Note: If the system is on Software Component SAP_UI release 754 SP16 or higher, follow Step 3.2.1. Otherwise, apply Step 3.2.2.

3.2.1 ALV Grid – Option 1

Transaction code	SE24
Class	CL_SALV_LOGGER
Method (under local definitions/implementations section)	if_salv_logger~log_task_begin

Create the SPI in the class mentioned below at the start of the given method by copying the code snippet.

Snippet:

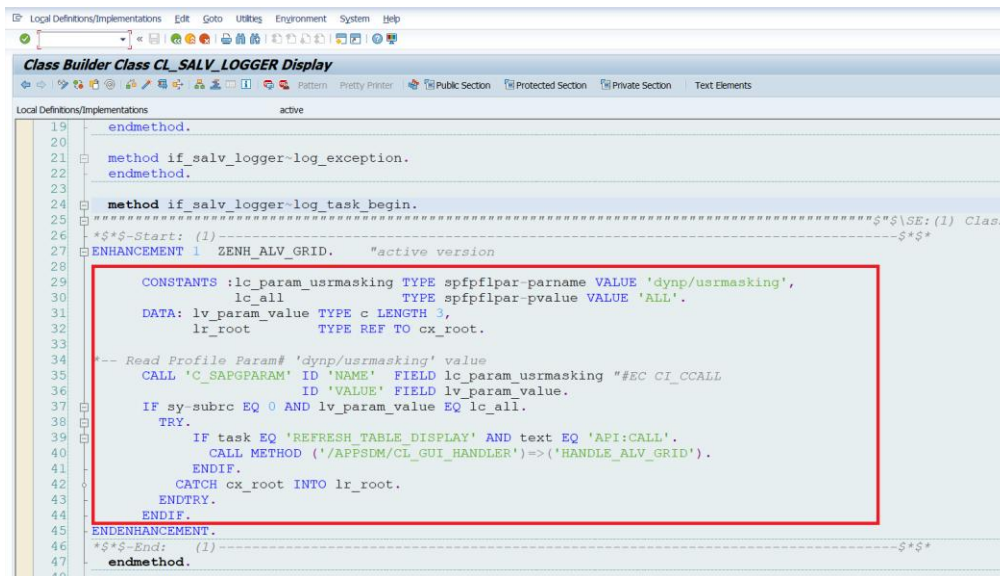
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  TRY.
    IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
    ENDIF.
    CATCH cx_root INTO lr_root.
  ENDTRY.
ENDIF.

```

Screenshot:



```

19: endmethod.
20:
21: method if_salv_logger-log_exception.
22: endmethod.
23:
24: method if_salv_logger-log_task_begin.
25:
26: *$*S-Start: (1)-----$*S\SE:(1) Class
27: ENHANCEMENT 1 ZENH_ALV_GRID. "active version
28:
29:     CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
30:                 lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
31:     DATA: lv_param_value TYPE c LENGTH 3,
32:            lr_root TYPE REF TO cx_root.
33:
34:     *-- Read Profile Param# 'dynp/usrmasking' value
35:     CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
36:          ID 'VALUE' FIELD lv_param_value.
37:     IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
38:     TRY.
39:         IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
40:             CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
41:         ENDIF.
42:         CATCH cx_root INTO lr_root.
43:     ENDTRY.
44:     ENDF.
45: ENDENHANCEMENT.
46: *$*S-End: (1)-----$*S*
47: endmethod.
48:

```

3.2.2 ALV Grid – Option 2

Transaction code	SE24
Class	CL_GUI_ALV_GRID
Method	REFRESH_TABLE_DISPLAY

Create the SPI in the below class at the start of the method by copying the code snippet.

Snippet:

```

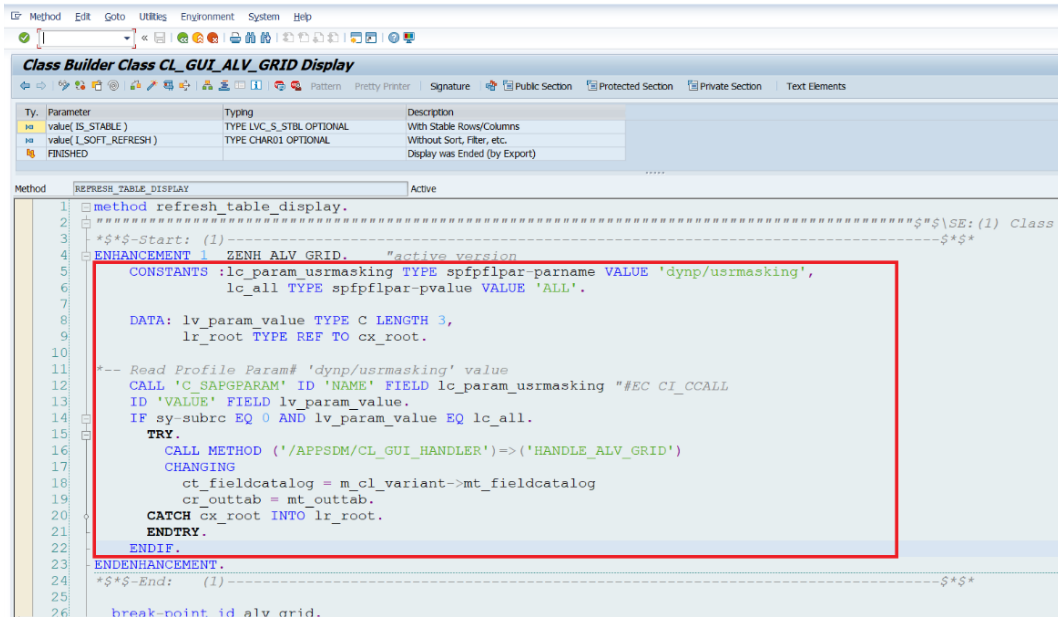
CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE c LENGTH 3,
      lr_root TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
TRY.
    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID')
    CHANGING
        ct_fieldcatalog = m_cl_variant->mt_fieldcatalog
        cr_outtab = mt_outtab.
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

Screenshot:



```

1  method refresh_table display.
2  *****
3  *$*-Start: (1)-----
4  ENHANCEMENT 1 ZENH_ALV_GRID.      "active version
5  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
6             lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
7
8  DATA: lv_param_value TYPE C LENGTH 3,
9         lr_root TYPE REF TO cx_root.
10
11
12  *-- Read Profile Param# 'dynp/usrmasking' value
13  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
14  ID 'VALUE' FIELD lv_param_value.
15  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
16      TRY.
17          CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID')
18              CHANGING
19                  ct_fieldcatalog = m_cl_variant->mt_fieldcatalog
20                  cr_outtab = mt_outtab.
21          CATCH cx_root INTO lr_root.
22      ENDTRY.
23  ENDIF.
24  ENHANCEMENT.
25  *$*-End: (1)-----
26  break-point id alv_grid.
  
```

3.3 SAP GUI: ALV List

Note: If the system is on and above S4/HANA 2023 follow Step 3.3.2. Otherwise, apply Step 3.2.1.

3.3.1 ALV List – Option 1

Transaction code	SE38
Program	LKKBLF01
Subroutine	FIRST_LIST_DISPLAY

Create the SPI in the below include program at the start of the subroutine by copying the snippet code below.

Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all                TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root type REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all and
   gt_stack-listtype eq 'S'.

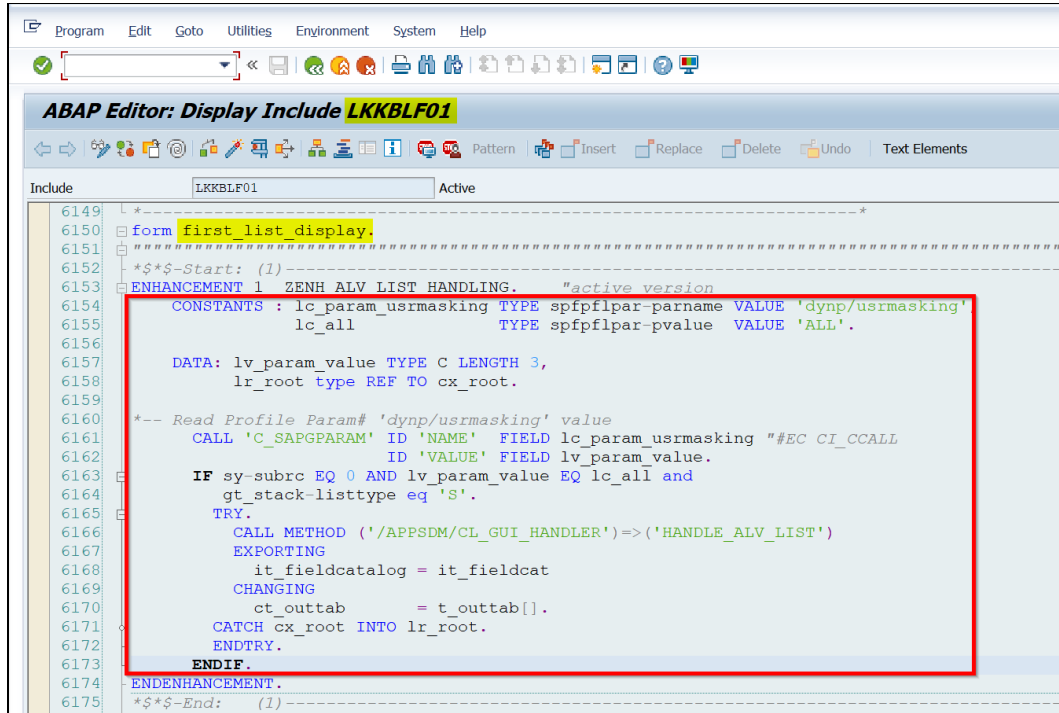
  TRY.

    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
    EXPORTING
      it_fieldcatalog = it_fieldcat
    CHANGING
      ct_outtab       = t_outtab[].

    CATCH cx_root INTO lr_root.
  ENDTRY.
ENDIF.

```

Screenshot:



```

6149  *-----*
6150  form first_list_display.
6151  *-----*
6152  *$*-Start: (1)-----*
6153  ENHANCEMENT 1 ZENH ALV LIST HANDLING. "active version
6154  CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking'
6155              lc_all              TYPE spfpflpar-pvalue  VALUE 'ALL'.
6156
6157  DATA: lv_param_value TYPE C LENGTH 3,
6158        lr_root type REF TO cx_root.
6159
6160  *-- Read Profile Param# 'dynp/usrmasking' value
6161  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
6162                      ID 'VALUE' FIELD lv_param_value.
6163  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all and
6164  gt_stack-listtype eq 'S'.
6165  TRY.
6166      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
6167      EXPORTING
6168          it_fieldcatalog = it_fieldcat
6169      CHANGING
6170          ct_outtab      = t_outtab[].
6171  CATCH cx_root INTO lr_root.
6172  ENDTRY.
6173  ENDIF.
6174  ENDENHANCEMENT.
6175  *$*-End: (1)-----*

```


3.3.2 ALV List – Option 2

Transaction code	SE24
Class	CL_SALV_LOGGER
Method (under local definitions/implementations section)	if_salv_logger~log_task_begin

Create the SPI in the class mentioned below at the start of the given method by copying the code snippet

Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value   TYPE c LENGTH 3,
      lr_root          TYPE REF TO cx_root,
      lc_fieldcat_list TYPE string VALUE '(SAPLKKBL)IT_FIELDCAT',
      lc_outtab_list   TYPE string VALUE '(SAPLKKBL)T_OUTTAB[]',
      lc_list_type     TYPE string VALUE '(SAPLKKBL)GT_STACK-LISTTYPE'.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                        ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    TRY.
        IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
            CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').

            ELSEIF task EQ 'FIRST_LIST_DISPLAY'.
*
                ASSIGN (lc_outtab_list) TO FIELD-SYMBOL(<fs_outtab_list>).
                ASSIGN (lc_fieldcat_list) TO FIELD-SYMBOL(<fs_fieldcat_list>).
                ASSIGN (lc_list_type) TO FIELD-SYMBOL(<fs_list_type>).
                IF <fs_list_type> IS ASSIGNED AND <fs_list_type> EQ 'S'.

                    IF <fs_outtab_list> IS ASSIGNED AND <fs_fieldcat_list> IS ASSIGNED.

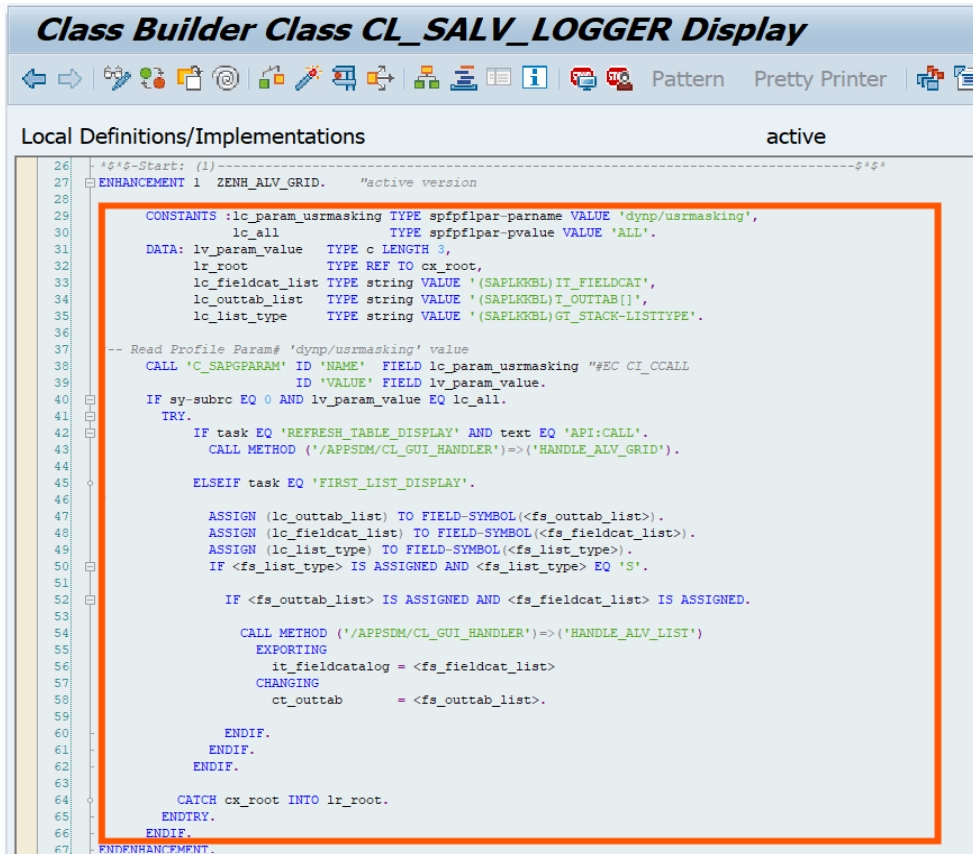
                        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
                        EXPORTING
                            it_fieldcatalog = <fs_fieldcat_list>
                        CHANGING
                            ct_outtab      = <fs_outtab_list>.

                        ENDIF.
                    ENDIF.
                ENDIF.

                CATCH cx_root INTO lr_root.
            ENDTRY.
        ENDIF.
    ENDIF.

```

Screenshot:



Class Builder Class CL_SALV_LOGGER Display

Local Definitions/Implementations active

```

26  *$*Start: (1)-----$*$
27  ENHANCEMENT 1 ZENH_ALV_GRID.      "active version
28
29  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
30             lc_all              TYPE spfpflpar-pvalue VALUE 'ALL'.
31  DATA: lv_param_value  TYPE c LENGTH 3,
32        lr_root         TYPE REF TO cx_root,
33        lc_fieldcat_list TYPE string VALUE '(SAPLKGBL)IT_FIELDCAT',
34        lc_outtab_list  TYPE string VALUE '(SAPLKGBL)T_OUTTAB[]',
35        lc_list_type    TYPE string VALUE '(SAPLKGBL)GT_STACK-LISTTYPE'.
36
37  -- Read Profile Param# 'dynp/usrmasking' value
38  CALL 'C_SARGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
39                  ID 'VALUE' FIELD lv_param_value.
40  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
41      TRY.
42          IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
43              CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
44
45          ELSEIF task EQ 'FIRST_LIST_DISPLAY'.
46
47              ASSIGN (lc_outtab_list) TO FIELD-SYMBOL(<fs_outtab_list>).
48              ASSIGN (lc_fieldcat_list) TO FIELD-SYMBOL(<fs_fieldcat_list>).
49              ASSIGN (lc_list_type) TO FIELD-SYMBOL(<fs_list_type>).
50              IF <fs_list_type> IS ASSIGNED AND <fs_list_type> EQ 'S'.
51
52                  IF <fs_outtab_list> IS ASSIGNED AND <fs_fieldcat_list> IS ASSIGNED.
53
54                      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
55                      EXPORTING
56                          it_fieldcatalog = <fs_fieldcat_list>
57                          CHANGING
58                              ct_outtab = <fs_outtab_list>.
59
60                      ENDIF.
61                  ENDIF.
62              ENDIF.
63
64              CATCH cx_root INTO lr_root.
65              ENDTRY.
66          ENDIF.
67  ENDENHANCEMENT.

```

3.4 SAP GUI: SE16 Standard List

Transaction code	SE38
Program	LSETBF01
Subroutine	TOP_OF_PAGE

Create the SPI in the include program at the start of the subroutine by copying the code snippet below.

Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all                TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root type REF TO cx_root.

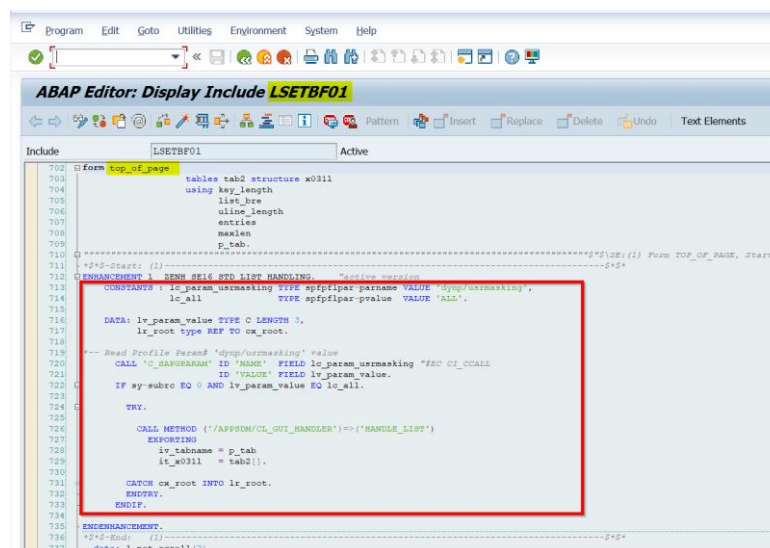
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                        ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.

    TRY.
        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_LIST')
            EXPORTING
                iv_tabname = p_tab
                it_x0311   = tab2[].

        CATCH cx_root INTO lr_root.
    ENDTRY.
ENDIF.

```

Screenshot:



3.5 SAP GUI: Search Helps (Begin)

Transaction code	SE38
Program	LSDSDF05
Subroutine	CALL_SHLP_EXIT

Create the SPI in the include program at the start of the subroutine by copying the code snippet below.

Snippet:

```

CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all        TYPE spfpflpar-pvalue  VALUE 'ALL'.

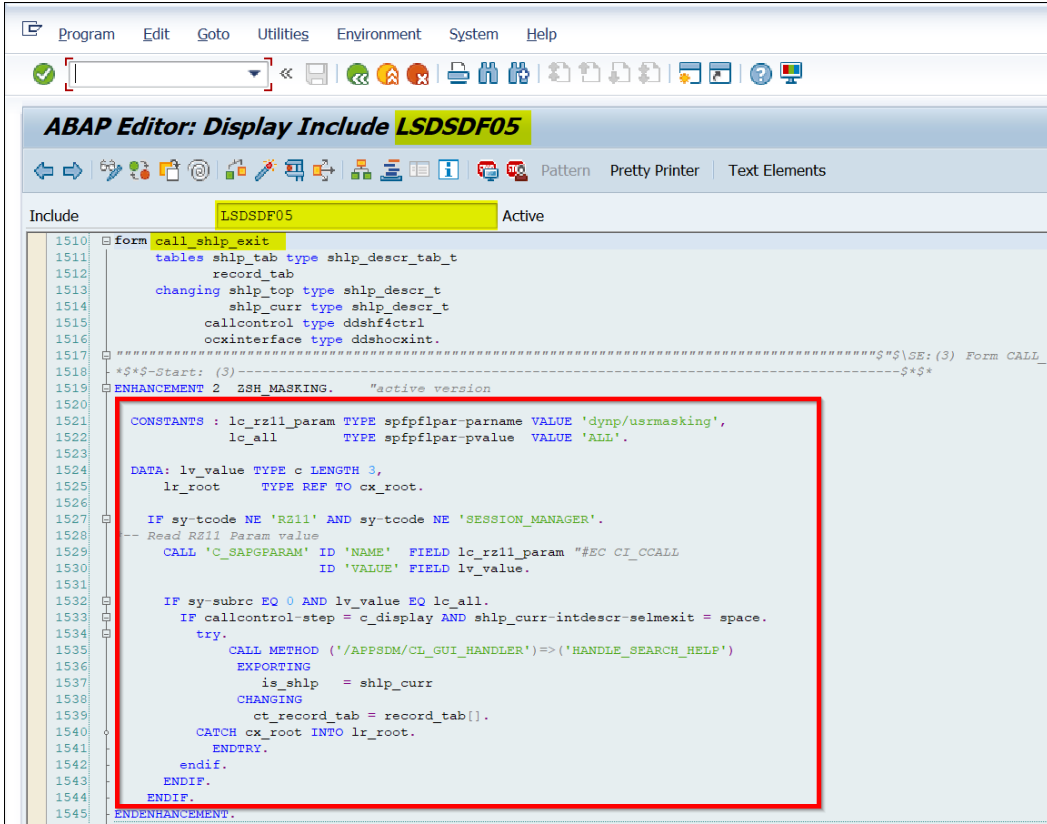
DATA: lv_value TYPE c LENGTH 3,
      lr_root   TYPE REF TO cx_root.

IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
  *-- Read RZ11 Param value
  CALL 'C_SAPGPARAM' ID 'NAME'  FIELD lc_rz11_param "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_value.

  IF sy-subrc EQ 0 AND lv_value EQ lc_all.
    IF callcontrol-step = c_display AND shlp_curr-intdescr-
selmexit = space.
      TRY.
        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
          EXPORTING
            is_shlp      = shlp_curr
          CHANGING
            ct_record_tab = record_tab[].
        CATCH cx_root INTO lr_root.
      ENDTRY.
    ENDIF.
  ENDIF.
ENDIF.

```

Screenshot:



```

1510 form call_shlp_exit
1511   tables shlp_tab type shlp_descr_tab_t
1512   record_tab
1513   changing shlp_top type shlp_descr_t
1514   shlp_curr type shlp_descr_t
1515   callcontrol type ddshfictrl
1516   cxcinterface type ddshockint.
1517   =====
1518   *$*Start: (3)-----
1519   ENHANCEMENT 2 ZSH_MASKING. "active version
1520
1521   CONSTANTS : lc_rz11_param TYPE spfplpar-parname VALUE 'dynp/usmasking',
1522             lc_all TYPE spfplpar-pvalue VALUE 'ALL'.
1523
1524   DATA: lv_value TYPE c LENGTH 3,
1525         lr_root TYPE REF TO cx_root.
1526
1527   IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
1528     -- Read RZ11 Param value
1529     CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
1530                      ID 'VALUE' FIELD lv_value.
1531
1532   IF sy-subrc EQ 0 AND lv_value EQ lc_all.
1533     IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit = space.
1534       try.
1535         CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
1536         EXPORTING
1537           is_shlp = shlp_curr
1538         CHANGING
1539           ct_record_tab = record_tab[].
1540       CATCH cx_root INTO lr_root.
1541     ENDTRY.
1542   endif.
1543   ENDIF.
1544   ENDIF.
1545   ENHENHANCEMENT.
  
```

3.6 SAP GUI: Search Helps (End)

Create the SPI in the include program at the end of the subroutine by copying the code snippet below.

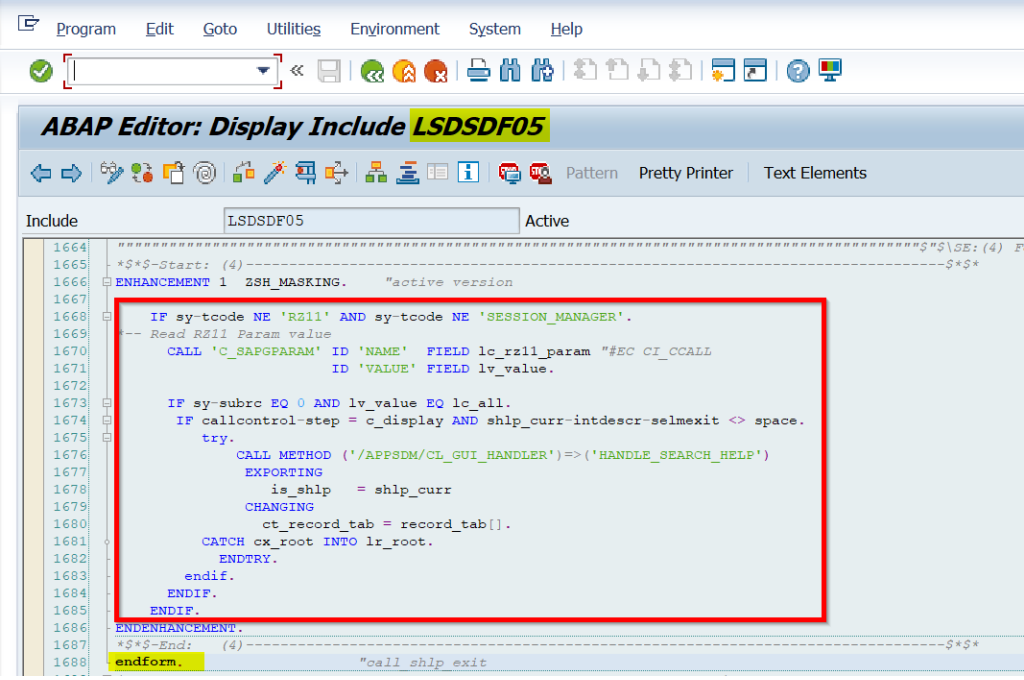
Snippet:

```

IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
*-- Read RZ11 Param value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_value.

IF sy-subrc EQ 0 AND lv_value EQ lc_all.
IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit <> space.
  try.
    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
    EXPORTING
      is_shlp = shlp_curr
    CHANGING
      ct_record_tab = record_tab[].
  CATCH cx_root INTO lr_root.
  ENDTRY.
ENDIF.
ENDIF.
ENDIF.
  
```

Screenshot:



The screenshot shows the ABAP Editor interface with the title bar "ABAP Editor: Display Include LSDSDF05". The menu bar includes Program, Edit, Goto, Utilities, Environment, System, and Help. The toolbar contains various icons for file operations and editing. The main window displays the code for the LSDSDF05 include file, which is currently active. A red rectangular box highlights a specific section of the code, which is as follows:

```

1664 *****
1665 *$*-Start: (4)-----
1666 ENHANCEMENT 1 ZSH_MASKING. "active version
1667
1668 IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
1669   Read RZ11 Param value
1670   CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EO CI_CCALL
1671   ID 'VALUE' FIELD lv_value.
1672
1673   IF sy-subrc EQ 0 AND lv_value EQ lc_all.
1674     IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit <> space.
1675       try.
1676         CALL METHOD ('/APPSPDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
1677         EXPORTING
1678           is_shlp = shlp_curr
1679         CHANGING
1680           ct_record_tab = record_tab[].
1681       CATCH cx_root INTO lr_root.
1682     ENDTRY.
1683   endif.
1684 ENDIF.
1685 ENDENHANCEMENT.
1686 *$*-End: (4)-----
1687 endform. "call shlp exit

```

3.7 SAP GUI: WRITE Statements

Transaction code	SE38
Program	SAPMSSY0
Subroutine	thew_list_masking2

Create the SPI via enhancement in the above-mentioned program at the start of the mentioned subroutine by copying the code snippet below.

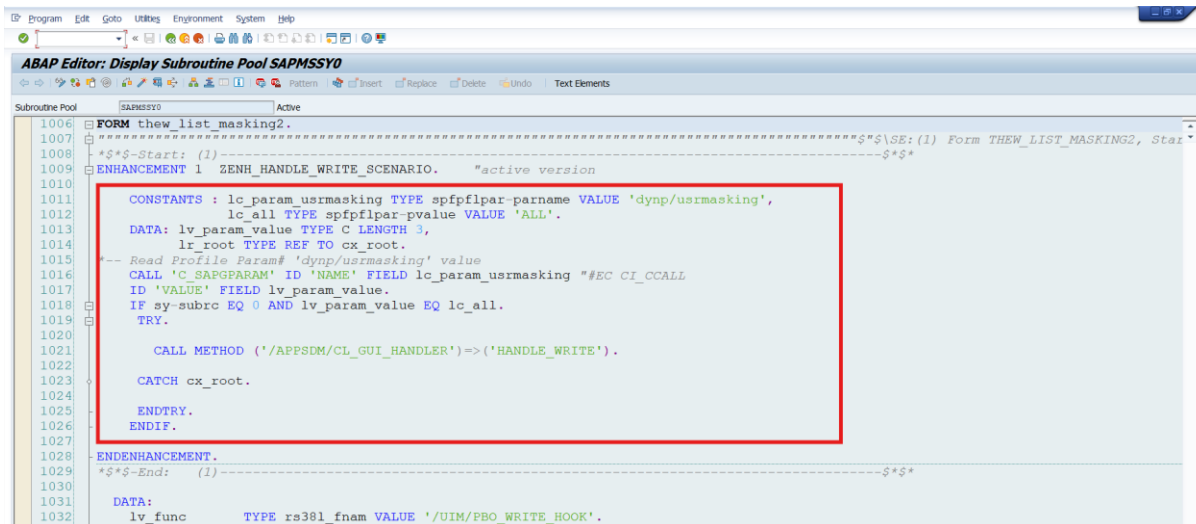
Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE C LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  TRY.
    CALL METHOD ('/APPSPDM/CL_GUI_HANDLER')=>('HANDLE_WRITE').
  CATCH cx_root.
  ENDTRY.
ENDIF.

```

Screenshot:



Note: Perform a syntax check before activating the code. Activating code with syntax errors can lead to system-wide runtime errors.

4 SAP UI5/Fiori (Enhancement / Modification)

4.1 SAP UI5/Fiori: Process Entity Set Data

Note:

If the system is on S4HANA 1709 and above, follow Step 4.1.1. Otherwise, apply Step 4.1.2 if SAP Product version is below S4HANA 1709 and on and above ECC 6.0 Ehp7.

4.1.1 Create the SPI in the class at the start of the method by copying the snippet code below

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	POST_PROCESS_READ_ENTITYSET

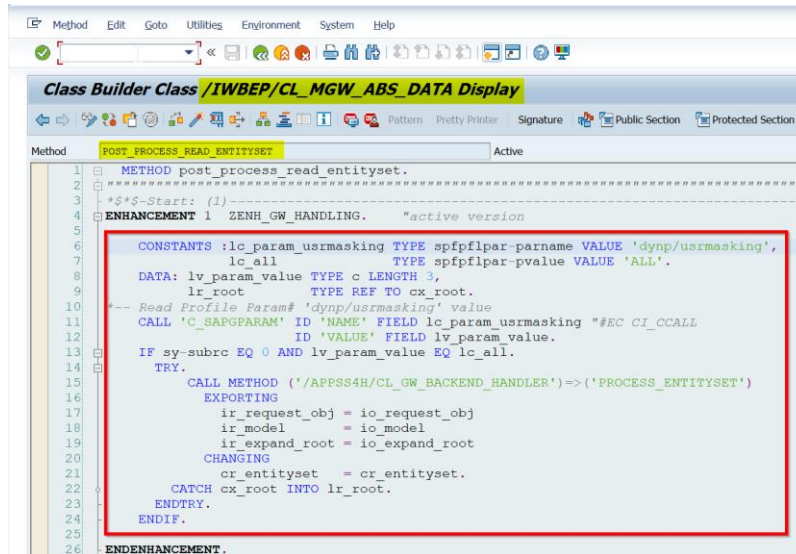
Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-
parname VALUE 'dynp/usrmasking',
lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
TRY.
CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
EXPORTING
ir_request_obj = io_request_obj
ir_model = io_model
ir_expand_root = io_expand_root
CHANGING
cr_entityset = cr_entityset.
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```


Screenshot:



```

1  METHOD post_process_read_entityset.
2  *****
3  *$*Start: (1)-----
4  ENHANCEMENT 1 ZENH_GW_HANDLING. "active version
5
6  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
7             lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
8  DATA: lv_param_value TYPE c LENGTH 3,
9         lr_root TYPE REF TO cx_root.
10 *-- Read Profile Param# 'dynp/usrmasking' value
11 CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
12 ID 'VALUE' FIELD lv_param_value.
13 IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
14 TRY.
15 CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
16 EXPORTING
17   ir_request_obj = io_request_obj
18   ir_model       = io_model
19   ir_expand_root = io_expand_root
20 CHANGING
21   cr_entityset = cr_entityset.
22 CATCH cx_root INTO lr_root.
23 ENDTRY.
24 ENDIF.
25
26 ENDENHANCEMENT.

```

4.1.2 Create the SPI in the class at the end of the method by copying the snippet code below

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	/IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITYSET

Snippet:

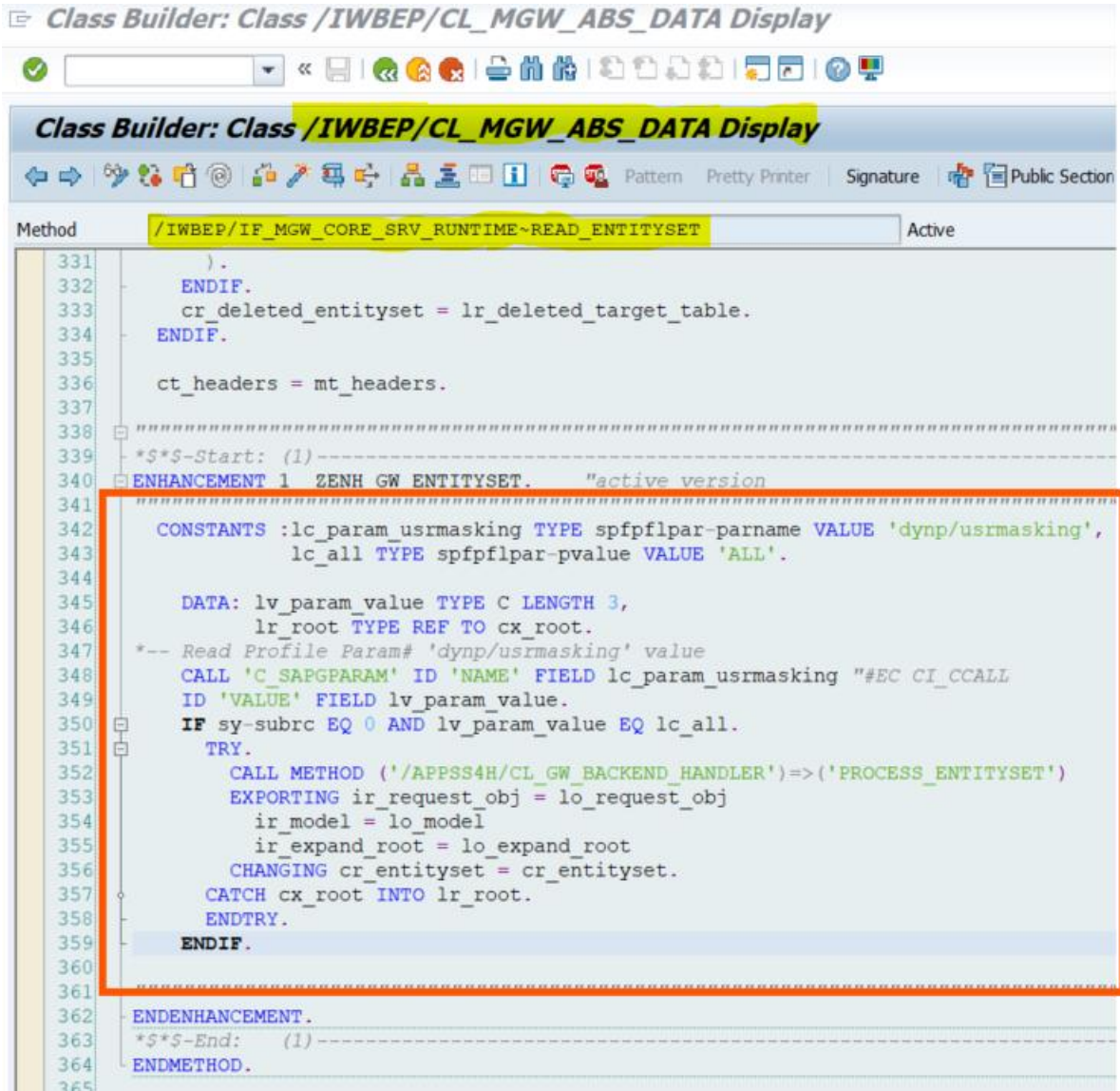
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE c LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
TRY.
CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
EXPORTING ir_request_obj = io_request_obj
          ir_model       = io_model
          ir_expand_root = io_expand_root
CHANGING cr_entityset = cr_entityset.
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

Screenshot:



```

Class Builder: Class /IWBEP/CL_MGW_ABS_DATA Display

Method /IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITYSET Active

331      ).
332      ENDIF.
333      cr_deleted_entityset = lr_deleted_target_table.
334      ENDIF.
335
336      ct_headers = mt_headers.
337
338      =====
339      *$*Start: (1)-----
340      ENHANCEMENT 1 ZENH GW ENTITYSET.      "active version
341
342      CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
343                  lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
344
345      DATA: lv_param_value TYPE C LENGTH 3,
346              lr_root TYPE REF TO cx_root.
347      *-- Read Profile Param# 'dynp/usrmasking' value
348      CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CALL
349      ID 'VALUE' FIELD lv_param_value.
350      IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
351      TRY.
352          CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
353          EXPORTING ir_request_obj = lo_request_obj
354                  ir_model = lo_model
355                  ir_expand_root = lo_expand_root
356          CHANGING cr_entityset = cr_entityset.
357          CATCH cx_root INTO lr_root.
358      ENDTRY.
359      ENDIF.
360
361      =====
362      ENDENHANCEMENT.
363      *$*End: (1)-----
364      ENDMETHOD.
365
  
```

4.2 SAP UI5/Fiori: Process entity data.

Note:

If the system is on S4HANA 1709 and above, follow Step 4.2.1. Otherwise, apply Step 4.2.2 if SAP Product version is below S4HANA 1709 and on and above ECC 6.0 Ehp7.

4.2.1 Create the SPI in the below-mentioned class at the start of method by copying the below snippet.

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	POST_PROCESS_READ_ENTITY

Snippet:

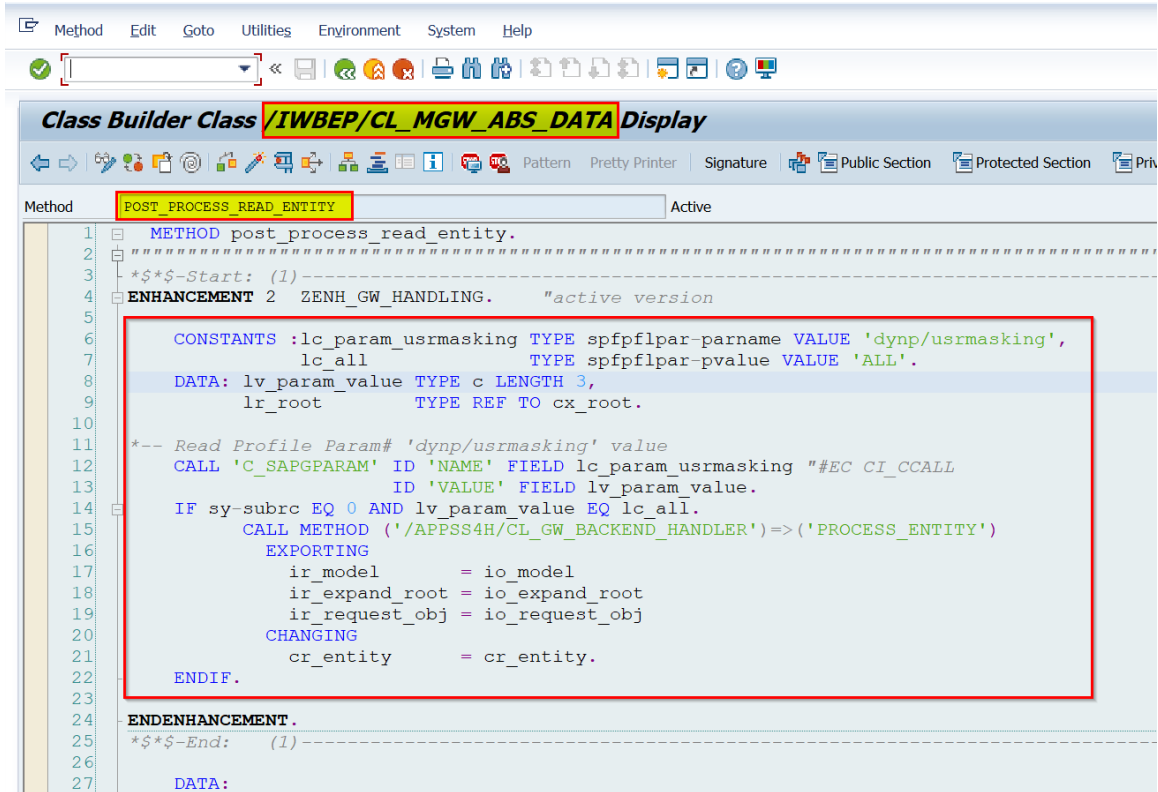
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
           ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
    EXPORTING
        ir_model          = io_model
        ir_expand_root    = io_expand_root
        ir_request_obj     = io_request_obj
    CHANGING
        cr_entity         = cr_entity.
ENDIF.

```

Screenshot:



```

Method: POST_PROCESS_READ_ENTITY Active

1  METHOD post_process_read_entity.
2  *****
3  *$*-$-Start: (1)-----
4  ENHANCEMENT 2 ZENH_GW_HANDLING. "active version
5
6  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
7             lc_all              TYPE spfpflpar-pvalue VALUE 'ALL'.
8  DATA: lv_param_value TYPE c LENGTH 3,
9         lr_root        TYPE REF TO cx_root.
10
11 *-- Read Profile Param# 'dynp/usrmasking' value
12 CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
13             ID 'VALUE' FIELD lv_param_value.
14 IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
15     CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
16     EXPORTING
17         ir_model      = io_model
18         ir_expand_root = io_expand_root
19         ir_request_obj = io_request_obj
20     CHANGING
21         cr_entity     = cr_entity.
22 ENDIF.
23
24 ENDENHANCEMENT.
25 *$*-$-End: (1)-----
26
27 DATA:

```

4.2.2 Create the SPI in the below-mentioned class at the end of method by copying the below snippet.

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	/IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITY

Snippet:

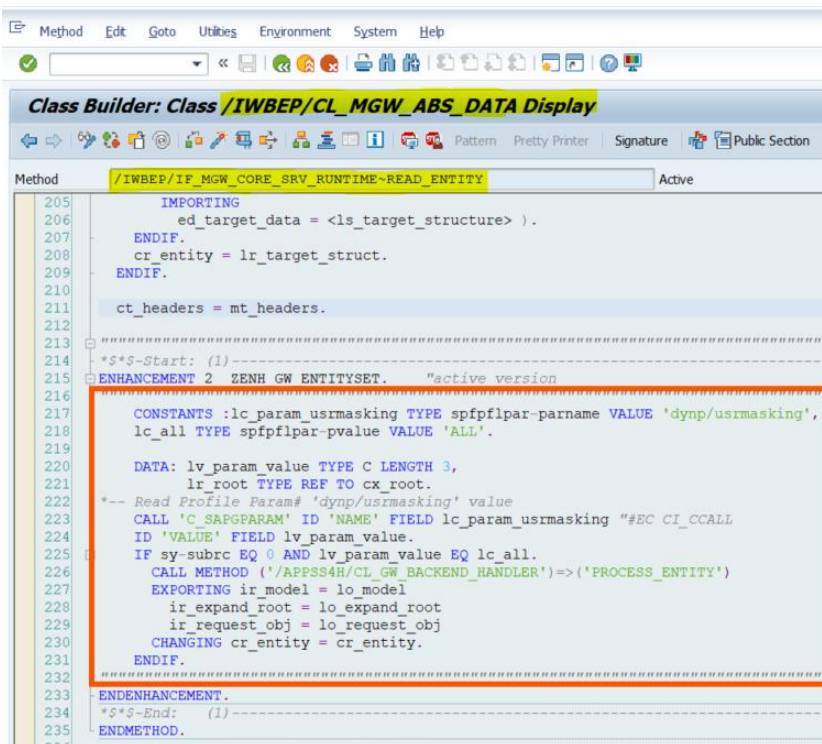
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
  EXPORTING ir_model = lo_model
            ir_expand_root = lo_expand_root
            ir_request_obj = lo_request_obj
  CHANGING cr_entity = cr_entity.
ENDIF.

```

Screenshot:



4.3 SAP UI5/Fiori: Update Scenario Processing Part 1

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	/IWBEP/IF_MGW_CORE_SRV_RUNTIME~UPDATE_ENTITY

Create the SPI in the class at the start of the method by copying the code snippet below.

Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_request_obj  TYPE REF TO /iwbe/CL_MGW_REQUEST,
      ls_request_context_abac TYPE
/iwbe/IF_MGW_CORE_SRV_RUNTIME=>ty_s_mgw_request_context,
      lr_root         TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.

* Create Technical Request Object
ls_request_context_abac = is_request_details.

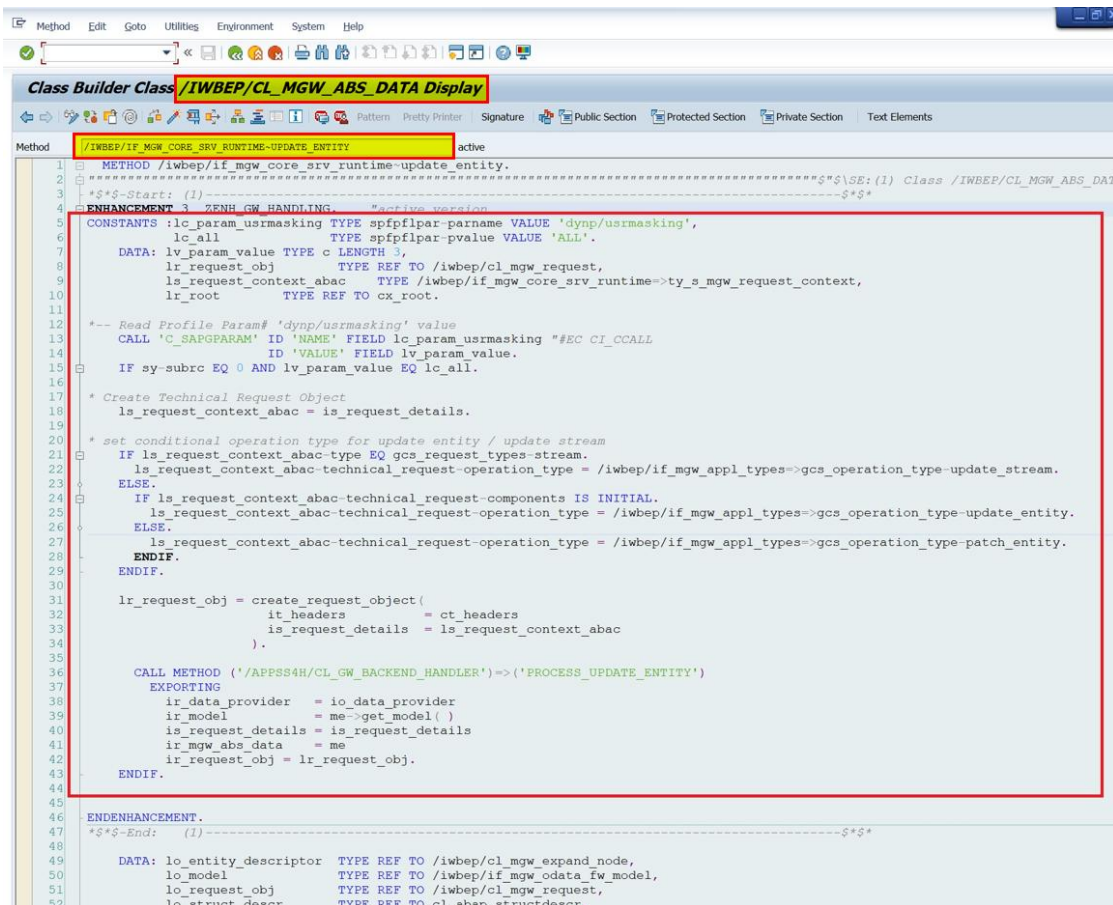
* set conditional operation type for update entity / update stream
IF ls_request_context_abac-type EQ gcs_request_types-stream.
  ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-update_stream.
ELSE.
  IF ls_request_context_abac-technical_request-components IS INITIAL.
    ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-update_entity.
  ELSE.
    ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-patch_entity.
  ENDIF.
ENDIF.

lr_request_obj = create_request_object(
    it_headers      = ct_headers
    is_request_details = ls_request_context_abac
).

CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_UPDATE_ENTITY')
EXPORTING
  ir_data_provider = io_data_provider
  ir_model          = me->get_model( )
  is_request_details = is_request_details
  ir_mgw_abs_data   = me
  ir_request_obj    = lr_request_obj.
ENDIF.

```

Screenshot:



```

Method: /IWBEP/IF_MGW_CORE_SRV_RUNTIME-UPDATE_ENTITY active

1 METHOD /iwbe/ifu_mgw_core_srv_runtime-update entity.
2 *****
3 *$*Start: (1)-----$*SE:(1) Class /IWBEP/CL_MGW_ABS_DATA
4 ENHANCEMENT 3 ZENH GW HANDLING. "active version
5
6 CONSTANTS :lc_param_usrmasking TYPE spfifpar-parname VALUE 'dynp/usrmasking',
7            lc_all TYPE spfifpar-pvalue VALUE 'ALL'.
8 DATA: lv_param_value TYPE c LENGTH 3,
9        lr_request_obj TYPE REF TO /iwbe/ifu_mgw_request,
10        ls_request_context_abac TYPE /iwbe/ifu_mgw_core_srv_runtime=>ty_s_mgw_request_context,
11        lr_root TYPE REF TO cx_root.
12
13 *-- Read Profile Param# 'dynp/usrmasking' value
14 CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
15            ID 'VALUE' FIELD lv_param_value.
16 IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
17
18 * Create Technical Request Object
19 ls_request_context_abac = is_request_details.
20
21 * set conditional operation type for update entity / update stream
22 IF ls_request_context_abac-type EQ gcs_request_types-stream.
23 ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_mgw_appl_types=>gcs_operation_type-update_stream.
24 ELSE.
25 IF ls_request_context_abac-technical_request-components IS INITIAL.
26 ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_mgw_appl_types=>gcs_operation_type-update_entity.
27 ELSE.
28 ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_mgw_appl_types=>gcs_operation_type-patch_entity.
29 ENDF.
30 ENDF.
31
32 lr_request_obj = create_request_object(
33     it_headers = ct_headers
34     is_request_details = ls_request_context_abac
35 ).
36
37 CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_UPDATE_ENTITY')
38 EXPORTING
39     ir_data_provider = io_data_provider
40     ir_model = me->get_model( )
41     is_request_details = is_request_details
42     ir_mgw_abs_data = me
43     lr_request_obj = lr_request_obj.
44 ENDF.
45
46 ENDENHANCEMENT.
47 *$*End: (1)-----$*
48
49 DATA: lo_entity_descriptor TYPE REF TO /iwbe/ifu_mgw_expand_node,
50        lo_model TYPE REF TO /iwbe/ifu_mgw_odata_fw_model,
51        lo_request_obj TYPE REF TO /iwbe/ifu_mgw_request,
52        lo_struct_descr TYPE REF TO cl_abap_structdescr.

```

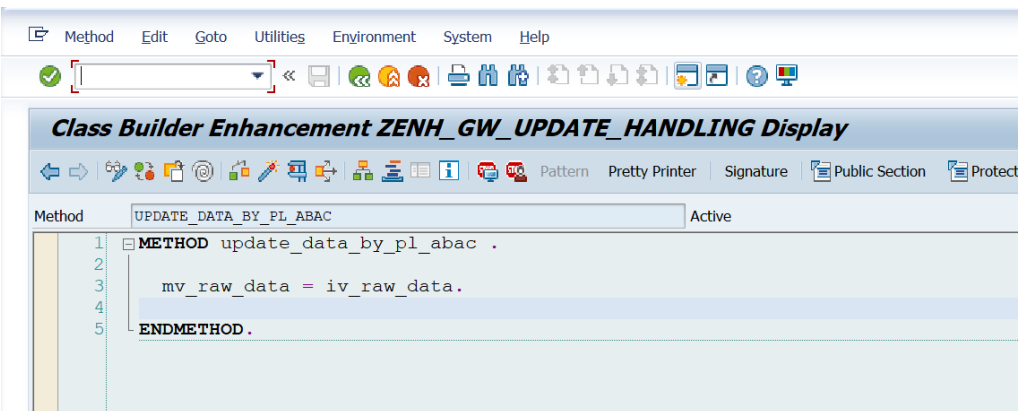
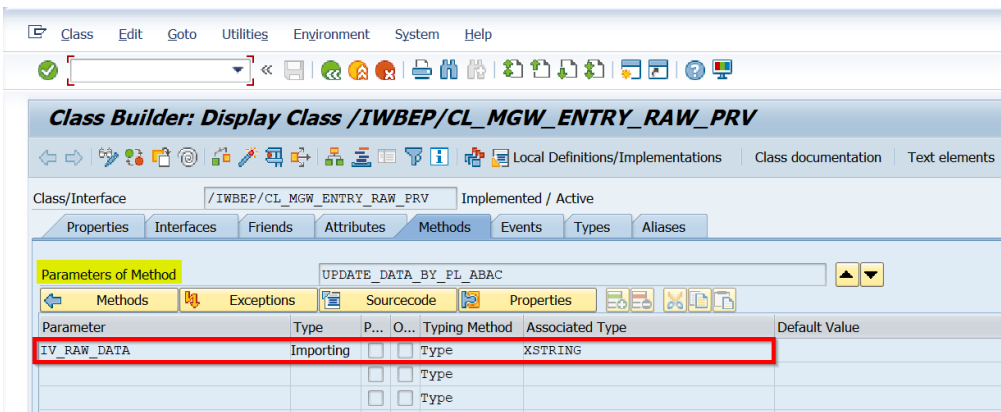
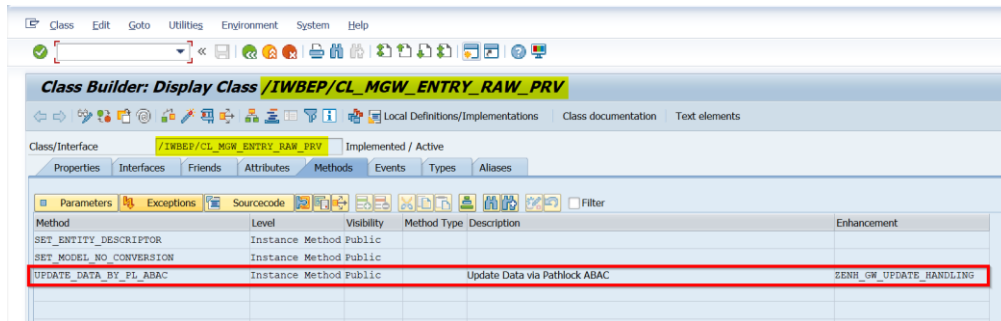
4.4 SAP UI5/Fiori: Update Scenario Processing Part 2

Transaction code	SE24
Class	/IWBEP/CL_MGW_ENTRY_RAW_PRV
New method	UPDATE_DATA_BY_PL_ABAC

Enhance the above class (/IWBEP/CL_MGW_ENTRY_RAW_PRV) to add a new method (UPDATE_DATA_BY_PL_ABAC). Add the following code to the newly created method.

Snippet:

```
mv_raw_data = iv_raw_data.
```


Screenshot:


4.5 SAP UI5/Fiori: Create Entity

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	/IWBEP/IF_MGW_CORE_SRV_RUNTIME~CREATE_ENTITY

Create the SPI in the aforementioned class at the end of the method by copying the code snippet below.

Snippet:

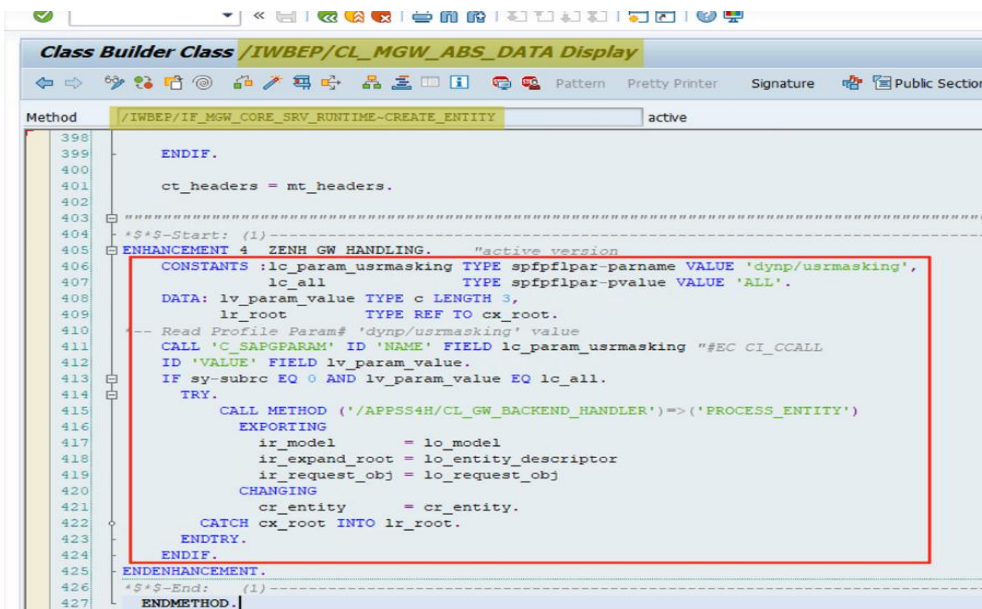
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    TRY.
        CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
        EXPORTING
            ir_model          = lo_model
            ir_expand_root    = lo_entity_descriptor
            ir_request_obj     = lo_request_obj
        CHANGING
            cr_entity         = cr_entity.
        CATCH cx_root INTO lr_root.
    ENDTRY.
ENDIF.

```

Screenshot:

Screenshot:



4.6 SAP UI5/Fiori: Prepare Response Header

Transaction code	SE24
Class	/IWBEP/CL_MGW_ABS_DATA
Method	POST_PROCESS_READ_ENTITYSET

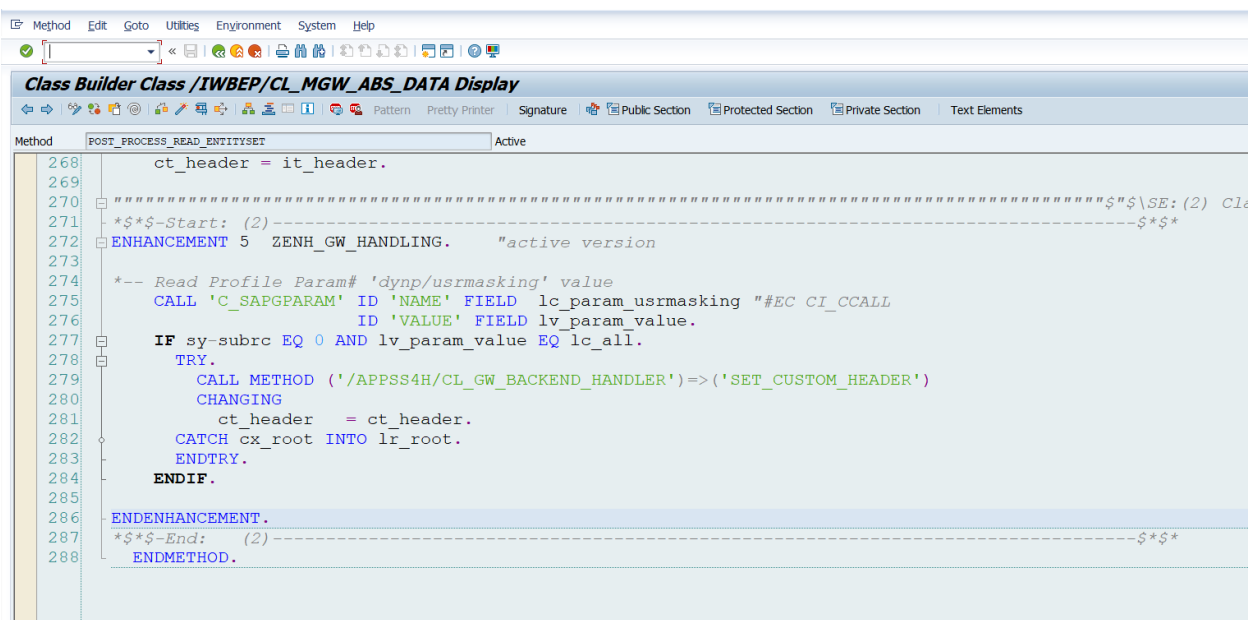
Create the SPI in the class at the end of the method by copying the code snippet below.

Snippet:

```

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                                ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    TRY.
        CALL
METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('SET_CUSTOM_HEADER')
        CHANGING
            ct_header = ct_header.
        CATCH cx_root INTO lr_root.
        ENDTRY.
    ENDIF.

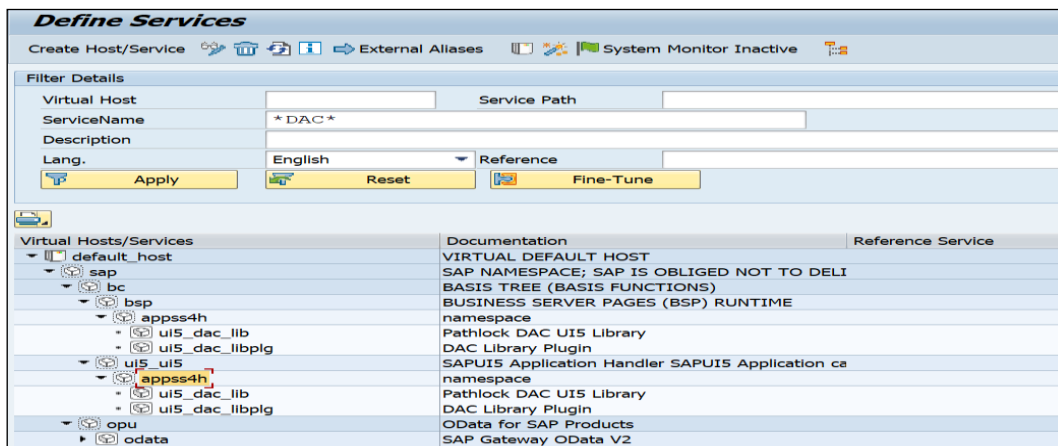
```



5 Activating Fiori Services (For UI5 Cases only)

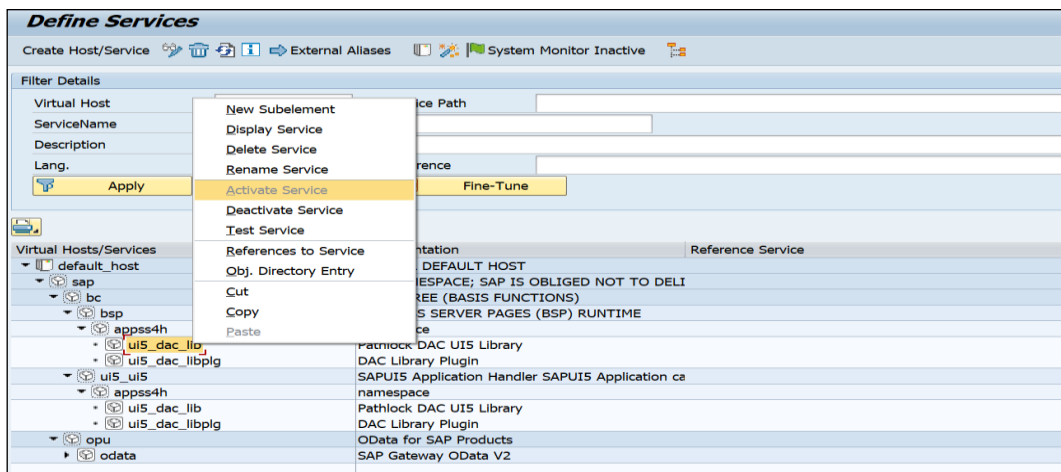
A) Activation of services

1. Run the ABAP T-code SICF.
2. Find the service node “/sap/bc/ui5_ui5/appss4h” and “/sap/bc/bsp /appss4h”.
3. Expand the “appss4h” node and activate all the services under it.



B) How to activate services?

1. Select the service name and click on it.
2. Once you click the context menu appears.
3. Click on “Activate Service” item in context menu.



- C) Assign the PFCG role **“/APPSS4H/DAC_BR_PLG”** to the user you wish to enable for the Pathlock UI5 solution.

Note: This role should be assigned to all users who have access to Fiori applications. If a user is not authorized, the absence of this role may lead to inconsistencies in their experience.

6 Frequently Asked Questions

1. What will be the impact, and how can we manage the changes related to upgrades, such as support pack upgrades or version upgrades?

Generally, upgrades do not have an impact because Single Points of Interception (SPIs) are dynamic calls primarily within the SAP_BASIS layer, which is not frequently updated. However, during upgrades, SAP may override these enhancements if SAP upgrades the same object. We provide a readiness report that indicates if an enhancement is not in place and prompts for re-implementation. This readiness report can also centrally monitor systems. Thus, if you have 20 systems where Pathlock ABAC is deployed, you can centrally monitor their status.

2. Is another approach available without code enhancement?

Code enhancements are the only method to achieve this in an SAP system without risking system integrity. We implement what we refer to as **Single Points of Interception**, which enables the application of **ABAC (Attribute-Based Access Control)** controls system-wide without modifying each individual application. This method is unique to our system and is not offered by other vendors without similar enhancements.

3. **What are the rollback procedures if we need to remove the tool from the system? For example, if it was implemented for years but we decide not to continue its use, or if it causes severe issues in the system and we need to revert the changes.**

If it becomes necessary to remove the tool from the system, we have established rollback procedures. These procedures leverage dynamic calls with embedded checks. In the event of any issues, a profile parameter accessible through RZ11 or the Basis console can effectively deactivate the tool. This parameter acts as a switch, turning off the tool and preventing any further impact on system performance or stability. Furthermore, if a decision is made to discontinue the tool's use after an extended period, it can be easily removed without causing downtime or disruptions to system operations.